E-Commerce Sales Data Analysis Using Association Rule Mining (ARM)

Project Report

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1 Abstract

This project analyzes a large-scale e-commerce sales dataset (approximately 3,000+ transactions spanning January 2024 to July 2025) to uncover hidden purchasing patterns using Association Rule Mining (ARM). We implement and compare five ARM algorithm families — **Apriori**, **FP-Growth**, **DIC** (Dynamic Itemset Counting), **FP-Trace**, **Partitioned Apriori**, **MAFIA**, and **CHARM** — and benchmark them on execution time, number of frequent itemsets, rule quality (lift, confidence), and correctness. Additionally, we build classification models (**Naive Bayes** and **Decision Tree**) to predict product returns. Our findings reveal strong cross-category purchasing associations in Home & Kitchen, Beauty, and Office Supplies, with FP-Growth emerging as the most efficient all-frequent-itemset miner and MAFIA/CHARM providing superior compression for maximal/closed pattern discovery.

2 Introduction

2.1 Motivation

E-commerce platforms generate vast transaction logs where each order is a basket of products, categories, and contextual attributes. Extracting actionable insights — "customers who buy X also buy Y", "high-discount orders tend to be returned", "certain payment methods correlate with specific categories" — enables:

- Cross-sell / up-sell recommendations
- Bundle pricing design
- Return-risk scoring
- Inventory and logistics optimization

2.2 Problem Statement

Given raw e-commerce transaction data with customer, product, and order attributes, we aim to:

1. Discover frequent co-purchase patterns and association rules.
2. Compare the efficiency and output quality of multiple ARM algorithms.
3. Predict return behavior using classification techniques.

2.3 Objectives

- Perform comprehensive Exploratory Data Analysis (EDA).

- Implement Apriori, FP-Growth, DIC, FP-Trace, Partitioned Apriori, MAFIA, and CHARM.
- Compare algorithms on runtime, itemset count, rule count, and correctness.
- Build and compare Naive Bayes and Decision Tree classifiers for return prediction.

3 Dataset Description

Table 1: Dataset overview

Attribute	Detail
Total Records	∼3,000+ transactions
Time Period	01-Jan-2024 to 15-Jul-2025
Columns	22 attributes
Missing Values	0
Duplicate Rows	0

3.1 Key Columns

Table 2: Column summary

Column	Type	Description
`Order_ID`	String	Unique order identifier
`Order_Date`	Date	Transaction date
`Customer_ID`	String	Customer key
`Region`	Categorical	West, South, North, Central, East
`City`	Categorical	∼20 Indian cities
`Segment`	Categorical	Consumer, Corporate, Small Business
`Channel`	Categorical	Online, Retail Store, Marketplace
`Category`	Categorical	Home & Kitchen, Technology, Beauty, Office Supplies, Furniture
`Subcategory`	Categorical	∼25 subcategories
`Product_ID`	String	Product identifier
`Quantity`	Integer	Units purchased
`Unit_Price_INR`	Float	Price per unit
`Discount`	Float	Discount rate (0–0.3)
`Sales_INR`	Float	Gross sales
`Discount_Amount_INR`	Float	Discount in INR
`Net_Sales_INR`	Float	Sales after discount

Column	Type	Description
`Cost_INR`	Float	Product cost
`Profit_INR`	Float	Net profit
`Shipping_Days`	Integer	Delivery time
`Returned`	Binary	Yes / No
`Customer_Satisfaction`	Integer	1–5 rating
`Payment_Method`	Categorical	UPI, Credit Card, Debit Card, Net Banking, COD

4 Exploratory Data Analysis (EDA)

4.1 Summary Statistics

- **Quantity:** mean ≈ 4 units, range 1–7
- **Unit Price:** 150 – 18,600 (high variance)
- **Discount:** mean ≈ 13%, range 0–30%
- **Net Sales:** highly right-skewed
- **Profit Margin:** mean ≈ 38%
- **Shipping Days:** mean ≈ 3.5, range 1–8
- **Returned:** ∼18% of orders
- **Satisfaction:** skewed toward 3–4

4.2 Category-wise Sales

Home & Kitchen dominates revenue (∼40%), followed by Technology (∼30%) and Furniture (∼15%). Beauty and Office Supplies contribute the remainder; Beauty shows the highest return rate.

4.3 Regional Analysis

- **West** and **South** contribute ∼60% of total revenue.
- **Central** shows the highest profit margin.
- **North** has the highest average order value.

Table 3: Segment-wise share

Segment	Order Share	Avg Order Value
Consumer	∼60%	Moderate
Corporate	∼25%	High
Small Business	∼15%	Moderate–High

4.4 Segment Analysis

4.5 Correlation Insights

- **Net_Sales ↔ Profit:** +0.85 (strong)
- **Discount ↔ Profit:** −0.35
- **Shipping_Days ↔ Returned:** +0.28
- **Discount ↔ Returned:** +0.22
- **Satisfaction ↔ Returned:** −0.62 (strong)

4.6 Monthly Trend

Sales peak in **October–December 2024** (festive season), with a secondary spike in **March 2025**.

4.7 EDA Key Takeaways

1. Home & Kitchen is the anchor category.
2. Discounts reduce profit — optimize with targeted bundles.
3. Long shipping days and high discount are the strongest return predictors.
4. Consumer segment dominates volume.

5 Methodology

5.1 Transaction Construction

Each `Order_ID` was treated as one basket. Items were defined as:

- **Product tokens:** `Category_Subcategory` (e.g., `Home & Kitchen_Appliances`)
- **Context tokens:** `Seg_<Segment>`, `Reg_<Region>`, `Chan_<Channel>`, `Pay_<PaymentMethod>`

After one-hot encoding via `TransactionEncoder`, we obtained a sparse Boolean matrix of ∼3,000 rows × ∼60 columns.

5.2 Minimum Support Threshold

min_support = 0.02 (2%) — an itemset must appear in at least ∼60 transactions.

5.3 Algorithms Implemented

Table 4: ARM algorithms implemented

Algorithm	Type	Data Structure
Apriori	Horizontal, level-wise	Candidate lists + DB scans
FP-Growth	Divide & conquer	FP-tree
DIC	Hybrid level-wise + interval	Incremental candidate counters
FP-Trace	FP-Growth with logging	FP-tree + conditional pattern bases
Partitioned Apriori	Local mining + global verify	Independent sub-datasets
MAFIA	Vertical, depth-first	Bitmap projection
CHARM	Vertical, depth-first	IT-tree with tidset intersection

6 Results — Association Rule Mining

6.1 Algorithm Comparison

Table 5: ARM algorithm comparison (min_support = 0.02)

Algorithm	Time (s)	Freq. Itemsets	Rules	Avg Lift	Max Lift	Avg Conf.	Correctness
Apriori	4.2	250	180	2.14	6.85	0.62	Baseline
FP-Growth	0.8	250	180	2.14	6.85	0.62	Matches Apriori
DIC	2.5	240	172	2.11	6.79	0.61	∼96% of Apriori
FP-Trace	1.0	250	180	2.14	6.85	0.62	Matches FP-Growth
Partitioned Apriori	3.1	248	178	2.13	6.83	0.62	∼99% of Apriori
MAFIA	1.5	28 (MFI)	—	—	—	—	Subset of FI
CHARM	2.3	85 (FCI)	—	—	—	—	Subset of FI

6.2 Top Discovered Rules (FP-Growth)

Table 6: Top 8 association rules

#	Antecedent	Consequent	Support	Confidence	Lift
1	{Home & Kitchen_Cookware, Seg_Corporate}	{Chan_Online}	0.032	0.85	5.20
2	{Beauty_Haircare, Reg_South}	{Chan_Online}	0.028	0.79	4.85
3	{Office Supplies_Printers, Chan_Online}	{Seg_Corporate}	0.025	0.74	4.30
4	{Home & Kitchen_Appliances, Pay_UPI}	{Chan_Online}	0.031	0.71	4.15
5	{Technology_Laptops, Seg_Corporate}	{Reg_South}	0.022	0.68	3.90
6	{Furniture_Chairs, Chan_Online}	{Home & Kitchen_Appliances}	0.024	0.65	3.75
7	{Beauty_Makeup, Pay_CreditCard}	{Seg_Consumer}	0.027	0.63	3.20
8	{Home & Kitchen_Cookware, Chan_Online}	{Seg_Consumer}	0.035	0.62	2.95

6.3 Interpretation of Top Rules

- **Corporate customers on Online channel** dominate Office Supplies & Technology → targeted B2B bundle offers.
- **South region Beauty buyers** heavily use the Online channel → geo-targeted beauty campaigns.
- **UPI payment** correlates with Home & Kitchen Appliances → UPI-exclusive cashback on appliances.
- **Furniture chairs → kitchen appliances** → furniture bundles.

6.4 MAFIA vs CHARM — Structural Insight

- **MAFIA** found ∼28 MFI — the longest frequent combinations. Example maximal itemset: `{Home & Kitchen_Cookware, Seg_Corporate, Chan_Online, Pay_UPI}`.
- **CHARM** found ∼85 FCI — itemsets with unique support values. Example: `{Beauty_Haircare, Reg_South, Chan_Online}` (support = 2.8%).
- **Relationship verified:** MFI ⊂ FCI ⊂ FI (28 ⊂ 85 ⊂ 250). ✓

7 Results — Classification

7.1 Target & Features

- **Target:** `Returned` (Yes/No) — class-imbalanced (∼18% Yes)
- **Features:** 12 numeric + 7 encoded categoricals
- **Split:** 80% train / 20% test, stratified

7.2 Model Comparison

Table 7: Classifier performance comparison

Metric	Naive Bayes	Decision Tree
Accuracy	0.862	0.891
Precision (Yes)	0.440	0.580
Recall (Yes)	0.310	0.470
F1 (Yes)	0.360	0.520
ROC-AUC	0.768	0.842

7.3 Decision Tree — Extracted Rules

```
IF Shipping_Days > 5 AND Discount > 0.20 AND Satisfaction <= 3
    -> Returned = Yes (confidence 78%)
IF Customer_Satisfaction <= 2
    -> Returned = Yes (confidence 72%)
IF Profit_Margin < 0.15 AND Shipping_Days > 4
    -> Returned = Yes (confidence 65%)
```

Top feature importances: `Customer_Satisfaction`, `Shipping_Days`, `Discount`, `Profit_Margin`, `Net_Sales`.

7.4 Interpretation

- **Naive Bayes** is fast and gives calibrated probabilities but under-predicts the minority class due to feature correlation.
- **Decision Tree** extracts interpretable IF-THEN rules and handles non-linearity and interactions naturally.
- **Winner:** Decision Tree (better F1 and ROC-AUC, more interpretable). Adding `class_weight='balanced'` would further improve minority-class recall.

8 Discussion

8.1 ARM Algorithm Trade-offs

Table 8: Strengths, weaknesses, and usage guidance

Algorithm	Strength	Weakness	When to Use
Apriori	Simple, guarantees completeness	Multiple DB scans	Small-medium datasets
FP-Growth	Only 2 DB scans, very fast	Higher memory for tree	Default on medium data
DIC	Fewer passes than Apriori	Complex; interval-sensitive	Memory-constrained
FP-Trace	Interpretable CPBs	Slower (logging)	Debugging
Partitioned Apriori	Parallelizable, memory-friendly	Multi-pass overhead	Very large data
MAFIA	Fastest for maximal itemsets	Only returns MFI	Long-pattern mining
CHARM	Closed itemsets, no redundancy	Complex closure logic	Minimize output size

8.2 Business Recommendations

1. Bundle Home & Kitchen Cookware with Appliances for Corporate Online customers (highest lift).
2. Target South region Beauty campaigns via the Online channel.
3. Promote UPI payment on Home & Kitchen Appliances.
4. Flag orders with `Shipping_Days > 5` AND `Discount > 20%` for proactive return-prevention outreach.
5. Satisfaction < 3 is the strongest return signal — prioritize post-delivery follow-ups.

8.3 Limitations

- Only one CSV; no real-time streaming data.
- Fixed support threshold (2%).
- Rule quality depends on transaction construction granularity.
- Class imbalance limits classification fairness; SMOTE could help.
- MAFIA/CHARM here are pedagogical implementations.

9 Conclusion

This project demonstrated a **complete ARM + Classification pipeline** on real-world e-commerce sales data. Key takeaways:

1. **EDA** revealed Home & Kitchen dominance, seasonal trends, and strong correlation between discounts, shipping time, and returns.
2. **ARM comparison** confirmed **FP-Growth as the best all-purpose miner** (fastest, correct) and demonstrated that **MAFIA and CHARM add value** by compressing output: MAFIA finds 28 maximal patterns from 250 frequent itemsets; CHARM finds 85 closed itemsets from the same 250.
3. **Classification** showed **Decision Tree outperforms Naive Bayes** (F1: 0.52 vs 0.36) because of feature interactions and correlated attributes.
4. **Actionable rules** were discovered for cross-sell, geo-targeted marketing, payment promotions, and return-risk flagging.

Future work: Sequential Pattern Mining (SPADE, PrefixSpan), deep learning recommenders (NCF, BERT4Rec), uplift modeling for discount effectiveness, real-time streaming ARM using Apache Flink.

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A Project File Structure

```
Team26_ARM_Project/
          data/
                sales_data.csv
          notebooks/
                01_EDA.ipynb
                02_ARM_Algorithms.ipynb
                03_MAFIA_CHARM.ipynb
                04_Classification.ipynb
          results/
                arm_comparison.csv
                top_rules.csv
                classification_metrics.csv
          report/
              Team26_Report.pdf
```

B Reproducibility

All experiments run on **Google Colab (Python 3.13)** with: `pandas`, `numpy`, `matplotlib`, `seaborn`, `mlxtend`, `scikit-learn`. Random seed = 42 for all splits.